CATHOLIC UNIVERSITY OF GHANA, FIAPRE

TOPIC:

The incidence of hypertension disorders it's associated risk factors, complications and preventive measures amongst pregnant women attending ANC at Daboya Community

Felayire Benedicta Jessica UGR0502415556

Aborigo Mary Atariguna UGR0502415766

Akansilum Vida Adaam UGR0502415562

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CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Pregnancy is widely recognized as a natural physiological process; however, it is also a critical period that exposes women to a range of health risks that may threaten the life of both the mother and the unborn child. Globally, maternal health remains a major public health concern, particularly in low- and middle-income countries where access to quality healthcare services is often limited. Among the numerous complications associated with pregnancy, hypertensive disorders of pregnancy (HDP) have been identified as one of the leading causes of maternal and perinatal morbidity and mortality worldwide.

Hypertensive disorders of pregnancy comprise a group of conditions including chronic hypertension, gestational hypertension, pre-eclampsia, and eclampsia. These conditions are generally characterized by elevated blood pressure (≥140/90 mmHg) and may be accompanied by proteinuria or organ dysfunction in severe cases (Bugri et al., 2023). These disorders can occur at different stages of pregnancy and may persist into the postpartum period if not properly managed.

Globally, hypertensive disorders complicate approximately 6–10% of all pregnancies, making them one of the most common medical complications during pregnancy (Bugri et al., 2023). More importantly, these disorders contribute significantly to maternal deaths. According to a World Health Organization (WHO) review, hypertensive disorders account for approximately 14% of all maternal deaths globally, highlighting their critical impact on maternal health outcomes.

Furthermore, hypertensive disorders particularly pre-eclampsia and eclampsia are responsible for over 70,000 maternal deaths and 500,000 fetal and neonatal deaths annually worldwide. This underscores the severe consequences of these conditions when they are not detected and managed early. The burden of these deaths is disproportionately higher in low- and middle-income countries, where approximately 99% of maternal deaths occur, with Sub-Saharan Africa alone accounting for nearly 64–66% of global maternal deaths.

In Sub-Saharan Africa, hypertensive disorders of pregnancy remain a major public health challenge due to limited access to antenatal care, late diagnosis, and inadequate management of complications. A systematic review and meta-analysis reported that the pooled prevalence of hypertensive disorders of pregnancy in the region is approximately 8%, with significant associated risks such as maternal mortality, preterm delivery, and low birth weight. Women with hypertensive disorders in this region are also 17 times more likely to die compared to normotensive pregnant women.

In addition, the burden of maternal mortality remains significantly higher in Sub-Saharan Africa compared to developed countries. The lifetime risk of maternal death in the region is estimated at 1 in 38, compared to 1 in 5,400 in high-income countries, reflecting stark inequalities in healthcare access and quality. These disparities further emphasize the importance of improving maternal healthcare services, particularly in rural and underserved communities.

In Ghana, maternal mortality continues to be a major health concern despite various interventions aimed at improving maternal health outcomes. The maternal mortality ratio in Ghana is estimated at 319 deaths per 100,000 live births, which is significantly higher than the global target recommended by the United Nations (WHO, 2022). Hypertensive disorders of pregnancy are among the leading causes of these deaths. Evidence indicates that severe pre-eclampsia and eclampsia account for approximately 16–23% of maternal deaths in Ghana.

Recent studies conducted in Ghana further highlight the prevalence of hypertensive disorders among pregnant women. For example, a study at Tamale Teaching Hospital reported a prevalence rate of 12.5%, indicating that the condition is still highly prevalent within the country (Bugri et al., 2023). Additionally, a review of maternal deaths in a major referral hospital in Ghana found that hypertensive disorders contributed to over 40% of maternal deaths, demonstrating the severity of the problem at the national level.

Despite these alarming statistics, much of the existing research in Ghana has focused on urban and tertiary healthcare facilities, leaving a significant gap in knowledge regarding rural communities such as Daboya. Rural areas often face unique challenges, including limited healthcare infrastructure, inadequate staffing, poor transportation systems, and low awareness of maternal health issues. These factors can lead to delayed diagnosis and management of hypertensive disorders, thereby increasing the risk of complications.

Moreover, socio-demographic factors such as low educational level, poverty, cultural beliefs, and limited access to health information further exacerbate the problem in rural settings. Many pregnant women may not recognize early warning signs of hypertensive disorders or may delay seeking care until complications arise. As a result, preventable maternal and neonatal deaths continue to occur.

Given this context, there is a critical need for localized studies to assess the incidence, risk factors, complications, and preventive measures associated with hypertensive disorders of pregnancy in rural communities like Daboya. Such studies will provide valuable data to inform targeted interventions, improve antenatal care services, and ultimately reduce maternal and neonatal mortality.

1.2 Problem Statement

Hypertensive disorders of pregnancy remain one of the most significant causes of maternal and neonatal morbidity and mortality worldwide. Despite advances in maternal healthcare, these conditions continue to pose serious health risks, particularly in low- and middle-income countries where healthcare systems are often under-resourced.

Globally, hypertensive disorders account for approximately 14% of maternal deaths, making them one of the leading causes of maternal mortality. In Sub-Saharan Africa, the burden is even greater, with hypertensive disorders contributing to approximately 16% of maternal deaths. These conditions are associated with severe complications such as preterm birth, low birth weight, organ failure, and stillbirth.

In Ghana, the situation is particularly concerning. With a maternal mortality ratio of 319 deaths per 100,000 live births, the country continues to face challenges in achieving global maternal health targets. Hypertensive disorders especially pre-eclampsia and eclampsia are among the leading direct causes of maternal deaths, accounting for up to 23% of maternal mortality cases.

Furthermore, hospital-based studies reveal a high prevalence of hypertensive disorders among pregnant women. For instance, the prevalence of hypertensive disorders at Tamale Teaching Hospital was reported to be 12.5%, indicating that a significant proportion of pregnant women are affected by these conditions (Bugri et al., 2023). Additionally, evidence from a major referral hospital shows that hypertensive disorders contribute to over 40% of maternal deaths, highlighting the severity of the issue.

Despite these findings, there is a lack of comprehensive data on the incidence and management of hypertensive disorders in rural communities such as Daboya. Most existing studies have been conducted in urban or tertiary healthcare settings, which may not accurately reflect the situation in rural areas. This gap in knowledge limits the ability of healthcare providers and policymakers to design effective interventions tailored to these communities.

Moreover, many pregnant women in rural areas have limited knowledge about the risk factors, complications, and preventive measures associated with hypertensive disorders. This lack of awareness often leads to late antenatal attendance, delayed diagnosis, and poor health-seeking behavior, thereby increasing the risk of adverse outcomes.

In addition, healthcare facilities in rural areas often face challenges such as inadequate medical equipment, shortage of skilled personnel, and limited access to diagnostic services. These challenges further hinder the early detection and effective management of hypertensive disorders in pregnancy.

Therefore, there is a pressing need to investigate the incidence, associated risk factors, complications, and preventive measures of hypertensive disorders among pregnant women attending antenatal care in Daboya Community. The findings of this study will help bridge the existing knowledge gap and provide evidence-based recommendations for improving maternal healthcare services in rural settings.

1.3 Objectives of the Study

The objectives of a research study provide a clear direction for the investigation by outlining what the researcher intends to achieve. In this study, the objectives are designed to explore the burden of hypertensive disorders of pregnancy (HDP), identify contributing factors, examine associated complications, and evaluate preventive practices among pregnant women attending antenatal care at Daboya Community.

1.3.1 General Objective

The general objective of this study is to assess the incidence of hypertensive disorders of pregnancy, associated risk factors, complications, and preventive measures among pregnant women attending antenatal care at Daboya Community.

This objective seeks to provide a comprehensive understanding of the magnitude of hypertensive disorders within the study area and to generate evidence that can inform clinical practice and public health interventions. By focusing on both clinical and behavioral aspects, the study aims to bridge the gap between disease occurrence and preventive healthcare practices.

1.3.2 Specific Objectives

To achieve the general objective, the study will pursue the following specific objectives:

1. To determine the incidence of hypertensive disorders among pregnant women attending antenatal care

This objective focuses on quantifying the proportion of pregnant women who develop hypertensive disorders within the study population. Understanding the incidence is crucial for assessing the burden of the condition in Daboya Community and comparing it with national and global statistics. Accurate incidence data will help healthcare providers evaluate whether hypertensive disorders are increasing, stable, or decreasing within the population.

Furthermore, determining incidence will assist in identifying high-risk groups and periods during pregnancy when women are most vulnerable. This information is essential for improving early detection strategies and strengthening antenatal care services (Gemechu et al., 2020).

2. To identify the risk factors associated with hypertensive disorders among pregnant women

This objective aims to examine the various maternal, socio-demographic, and clinical factors that predispose pregnant women to hypertensive disorders. These factors may include maternal age, parity, obesity, family history of hypertension, pre-existing medical conditions, and lifestyle behaviors such as diet and physical activity.

Identifying these risk factors is critical for developing targeted interventions and preventive strategies. For instance, women with known risk factors can be monitored more closely during antenatal visits. In addition, understanding the influence of socio-economic factors such as education level, income, and access to healthcare services will provide insight into the broader determinants of maternal health (WHO, 2022).

3. To assess the complications associated with hypertensive disorders in pregnancy

This objective seeks to evaluate both maternal and fetal complications resulting from hypertensive disorders of pregnancy. Maternal complications may include organ damage, stroke, and maternal death, while fetal complications may include intrauterine growth restriction, preterm birth, low birth weight, and stillbirth.

Assessing these complications is essential for understanding the severity and consequences of hypertensive disorders. It also helps to highlight the importance of early diagnosis and effective management. By documenting the types and frequency of complications within the study population, healthcare providers can improve clinical decision-making and prioritize high-risk cases (Bugri et al., 2023).

4. To examine the preventive measures adopted by pregnant women to reduce the risk of hypertensive disorders

This objective focuses on exploring the preventive practices employed by pregnant women, including regular antenatal attendance, dietary modifications, physical activity, and adherence to medical advice.

Understanding these preventive measures is important because many complications associated with hypertensive disorders can be avoided through early detection and appropriate interventions. This objective will also assess the level of awareness among pregnant women regarding hypertensive disorders and their prevention.

Additionally, the findings will help identify gaps in health education and inform strategies for improving maternal health awareness programs in the community.

1.4 Research Questions

Research questions are formulated to guide the study and provide a framework for data collection and analysis. They are directly derived from the research objectives and are designed to address the key issues under investigation.

The following research questions will guide this study:

1. What is the incidence of hypertensive disorders among pregnant women attending antenatal care at Daboya Community?

This question seeks to determine how widespread hypertensive disorders are within the study population. It will provide quantitative data that reflects the magnitude of the problem and allow for comparison with findings from other regions.

2. What are the risk factors associated with hypertensive disorders among pregnant women?

This question aims to identify the underlying causes and contributing factors that increase the likelihood of developing hypertensive disorders. It will explore both medical and socio-demographic determinants, providing a holistic understanding of the condition.

3. What complications are associated with hypertensive disorders in pregnancy?

This question focuses on the outcomes of hypertensive disorders, particularly the adverse effects on maternal and fetal health. The findings will highlight the severity of the condition and reinforce the importance of early intervention.

4. What preventive measures are used by pregnant women to reduce the risk of hypertensive disorders in pregnancy?

This question seeks to assess the knowledge, attitudes, and practices of pregnant women regarding the prevention of hypertensive disorders. It will help determine whether current health education efforts are effective and identify areas for improvement.

1.5 Significance of the Study

The significance of this study lies in its potential to contribute to improving maternal and neonatal health outcomes, particularly in rural communities such as Daboya. Hypertensive disorders of pregnancy remain a major public health challenge, and addressing this issue requires evidence-based interventions tailored to specific populations.

Firstly, the findings of this study will provide valuable data on the incidence of hypertensive disorders in Daboya Community. This information will help healthcare providers understand the magnitude of the problem within the local context and compare it with national and global trends. Such data are essential for effective planning and allocation of healthcare resources.

Secondly, the study will identify the key risk factors associated with hypertensive disorders among pregnant women. This will enable healthcare professionals to identify high-risk individuals early and provide targeted care and monitoring. Early identification of risk factors is critical for preventing the progression of the condition and reducing complications.

Thirdly, the study will contribute to improving clinical practice by providing insights into the complications associated with hypertensive disorders. Understanding these complications will help healthcare providers develop better management strategies and improve patient outcomes. It will also emphasize the importance of early diagnosis and timely intervention.

In addition, the study will assess the preventive measures adopted by pregnant women, thereby highlighting gaps in knowledge and practice. The findings will be useful for designing effective health education programs aimed at increasing awareness of hypertensive disorders and promoting healthy behaviors among pregnant women.

Furthermore, the study will benefit policymakers and public health authorities by providing evidence that can inform the development of maternal health policies and programs. It will also support efforts to achieve national and global targets for reducing maternal mortality, including the Sustainable Development Goals (SDGs), particularly Goal 3, which focuses on ensuring healthy lives and promoting well-being for all.

Academically, this study will contribute to the existing body of knowledge on hypertensive disorders of pregnancy, especially in rural settings where data are limited. It will serve as a reference for future researchers who may wish to conduct similar studies or build upon the findings.

Finally, the study will empower pregnant women by increasing awareness of the risks associated with hypertensive disorders and the importance of regular antenatal care. Improved awareness can lead to better health-seeking behavior, early detection of complications, and ultimately improved maternal and neonatal outcomes.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of existing literature on hypertensive disorders of pregnancy (HDP), with particular emphasis on incidence, associated risk factors, complications, and preventive measures. The review is organized according to the specific objectives of the study and draws on global, regional, and Ghanaian research to provide a broad and contextual understanding of the subject.

Hypertensive disorders of pregnancy remain a major contributor to maternal and neonatal morbidity and mortality worldwide. Over the years, several studies have examined different aspects of these disorders, including their prevalence, determinants, outcomes, and management strategies. However, much of the available literature is concentrated in urban and tertiary healthcare settings, leaving a gap in knowledge regarding rural communities.

This chapter therefore not only synthesizes existing research but also critically examines similarities and differences across studies, identifies gaps in knowledge, and establishes the relevance of the present study within the context of Daboya Community.

2.2 Incidence of Hypertensive Disorders in Pregnancy

The incidence of hypertensive disorders of pregnancy has been widely studied across different regions of the world, with findings indicating that the condition remains a significant public health concern. Globally, hypertensive disorders are estimated to complicate between 5% and 10% of all pregnancies, making them one of the most common medical complications during pregnancy (World Health Organization [WHO], 2022). Despite this global estimate, the incidence varies considerably across regions due to differences in healthcare systems, socio-economic conditions, and study methodologies.

In high-income countries, the incidence of hypertensive disorders tends to be relatively lower, largely due to improved access to quality antenatal care, early detection, and effective management of pregnancy-related complications. For instance, studies conducted in Europe and North America report prevalence rates ranging from 3% to 5%, reflecting the effectiveness of well-established maternal healthcare systems (American College of Obstetricians and Gynecologists [ACOG], 2020).

In contrast, developing regions, particularly Sub-Saharan Africa, experience a higher burden of hypertensive disorders of pregnancy. A systematic review and meta-analysis conducted by Gemechu et al. (2020) reported a pooled prevalence of approximately 8% in Sub-Saharan Africa. However, the study also noted that the actual burden may be underestimated due to limited diagnostic capacity and underreporting, especially in rural areas where access to healthcare services is restricted.

Country-specific studies within Africa further illustrate the variation in incidence. For example, research conducted in Ethiopia reported a prevalence of 9.3%, while studies in Nigeria have documented rates ranging from 7% to 12%, depending on the population and setting. These variations highlight the influence of contextual factors such as maternal education, nutritional status, and healthcare accessibility.

In Ghana, several hospital-based studies have documented the incidence of hypertensive disorders among pregnant women. Bugri et al. (2023) reported a prevalence of 12.5% at Tamale

Teaching Hospital, indicating a relatively high burden within the country. Other studies conducted in major referral hospitals in Accra and Kumasi have reported prevalence rates ranging between 6% and 13%, suggesting that hypertensive disorders remain a persistent challenge in Ghana's healthcare system.

However, it is important to note that most of these studies are conducted in urban or tertiary healthcare facilities, which may not accurately represent the situation in rural communities. Women in rural areas often face barriers such as poor transportation, limited healthcare infrastructure, and low awareness of maternal health issues, which can result in delayed diagnosis and underreporting of cases. Consequently, the true incidence of hypertensive disorders in rural settings such as Daboya may differ significantly from reported hospital-based figures.

Critically, while existing studies provide valuable insights into the prevalence of hypertensive disorders, there is a lack of community-based research that captures the experiences of women who may not access formal healthcare services. This gap underscores the need for localized studies that can provide a more accurate and comprehensive understanding of the incidence of hypertensive disorders in underserved communities.

2.3 Risk Factors Associated with Hypertensive Disorders

The development of hypertensive disorders of pregnancy is influenced by a wide range of risk factors, which can be broadly categorized into biological, medical, and socio-demographic factors. Understanding these risk factors is essential for early identification and prevention of the condition.

Maternal age has been consistently identified as a significant risk factor. Studies indicate that women at the extremes of reproductive age, particularly those below 20 years and above 35 years, are at increased risk of developing hypertensive disorders (WHO, 2022). Advanced maternal age is often associated with reduced vascular elasticity and increased susceptibility to cardiovascular complications, while younger mothers may lack the physiological maturity required to adapt to the demands of pregnancy.

Parity also plays a crucial role in the development of hypertensive disorders. Primigravida women, or those experiencing their first pregnancy, are more likely to develop conditions such as pre-eclampsia compared to multigravida women. This has been attributed to immunological factors and the body's initial response to pregnancy-related changes (Gemechu et al., 2020). However, some studies suggest that high parity may also be associated with increased risk due to cumulative physiological stress, indicating that the relationship between parity and hypertensive disorders is complex.

Obesity and lifestyle-related factors are increasingly recognized as important contributors to hypertensive disorders, particularly in urban populations. Women with a high body mass index are at greater risk due to metabolic and cardiovascular changes associated with excess body weight (ACOG, 2020). Poor dietary habits, physical inactivity, and excessive weight gain during pregnancy further exacerbate this risk.

Medical history is another critical determinant. Women with pre-existing conditions such as chronic hypertension, diabetes, or renal disease are more likely to develop hypertensive disorders during pregnancy. Additionally, a family history of hypertension suggests a genetic predisposition, further increasing the likelihood of developing the condition (WHO, 2022).

Socioeconomic factors also play a significant role in shaping maternal health outcomes. Women with low levels of education and income are less likely to access quality antenatal care, leading to delayed diagnosis and poor management of hypertensive disorders. Studies conducted in Sub-Saharan Africa have demonstrated a strong association between low socioeconomic status and increased risk of hypertensive disorders (Gemechu et al., 2020).

Critically, while many studies have focused on clinical and biological risk factors, there is limited attention given to cultural and behavioral influences, particularly in rural communities. Factors such as traditional beliefs, health-seeking behavior, and access to health information can significantly influence the development and management of hypertensive disorders. This highlights the need for context-specific research that incorporates both biomedical and socio-cultural perspectives.

2.4 Complications Associated with Hypertensive Disorders

Hypertensive disorders of pregnancy are associated with a wide range of complications that can affect both the mother and the fetus. These complications vary in severity depending on the type of hypertensive disorder, the timing of onset, and the quality of medical care received.

Maternal complications are often severe and can be life-threatening. Conditions such as pre-eclampsia and eclampsia can lead to seizures, stroke, and multi-organ failure if not properly managed. According to the World Health Organization (2022), hypertensive disorders account for approximately 14% of maternal deaths globally. In Ghana, these disorders contribute significantly to maternal mortality, with estimates ranging between 16% and 23% of maternal deaths.

In addition to direct maternal complications, hypertensive disorders can also lead to long-term health consequences, including chronic hypertension and cardiovascular disease later in life. This underscores the importance of early detection and effective management not only during pregnancy but also in the postpartum period.

Fetal and neonatal complications are equally significant. Hypertensive disorders can impair placental function, leading to reduced blood flow to the fetus. This can result in intrauterine growth restriction, low birth weight, and preterm delivery. In severe cases, it may lead to stillbirth or neonatal death. Gemechu et al. (2020) reported that hypertensive disorders are strongly associated with adverse perinatal outcomes, particularly in low-resource settings.

The severity of these complications is often influenced by the availability and quality of healthcare services. In settings where antenatal care is readily accessible, early detection and management can significantly reduce the risk of complications. However, in rural areas where healthcare resources are limited, delays in diagnosis and treatment contribute to higher rates of adverse outcomes.

A critical analysis of the literature reveals that while the complications of hypertensive disorders are well documented, there is limited research on how these complications are experienced and managed in rural communities. Most studies focus on hospital-based populations, which may not reflect the realities of women who have limited access to healthcare services. This highlights the need for studies that explore complications within community settings.

2.5 Conclusion

The literature reviewed in this chapter demonstrates that hypertensive disorders of pregnancy remain a major public health challenge globally, with a particularly high burden in Sub-Saharan Africa and Ghana. The incidence of these disorders varies across regions but is generally higher in low-resource settings due to limited access to healthcare and poor maternal health conditions.

Several risk factors have been identified, including maternal age, parity, obesity, medical history, and socioeconomic status. These factors interact in complex ways, influencing the likelihood of developing hypertensive disorders and the severity of outcomes.

The complications associated with hypertensive disorders are severe and can lead to both maternal and neonatal deaths if not properly managed. Although effective interventions exist, their implementation is often limited in rural areas due to various structural and socio-cultural barriers.

Importantly, the review highlights a significant gap in research focusing on rural communities such as Daboya. Most existing studies are hospital-based and may not accurately represent the experiences of women in underserved areas. This gap justifies the need for the current study, which aims to provide localized data and contribute to improving maternal health outcomes in the study area.

CHAPTER THREE

METHODOLOGY

3.1 Background of the Study

The study will be conducted in Daboya, a rural community located in the Savannah Region of Ghana. Daboya serves as the district capital of the North Gonja District and plays a significant role in the administrative and socio-economic activities within the area. The community is predominantly rural, with most residents engaged in subsistence farming, fishing, and small-scale trading, which are highly dependent on seasonal rainfall patterns (Ghana Statistical Service [GSS], 2021). These economic conditions influence household income levels and access to healthcare services.

The population of Daboya comprises diverse ethnic groups, including the Gonja and Dagomba, among others. Cultural beliefs and traditional practices remain influential in shaping health-seeking behaviors, particularly among women. In many rural Ghanaian communities, pregnancy-related conditions are sometimes initially managed through traditional practices or by traditional birth attendants before seeking formal healthcare services, which can delay early diagnosis and treatment of complications (Ganle et al., 2016).

Healthcare delivery in Daboya is provided through a district hospital and several primary healthcare facilities, including Community-based Health Planning and Services (CHPS) compounds. These facilities offer essential maternal health services such as antenatal care, skilled delivery, and postnatal care. Despite these services, challenges such as inadequate medical equipment, shortage of skilled health personnel, and inconsistent supply of essential medicines continue to affect the quality of care (Ministry of Health Ghana, 2020).

Accessibility to healthcare services remains a major concern, as many communities are located far from health facilities, and poor road networks further limit access, especially during the rainy season. This often results in delayed or missed antenatal visits, which are critical for early detection of complications such as hypertensive disorders (World Health Organization [WHO], 2022).

Antenatal care services in the community are designed to monitor pregnancy progression, detect complications early, and provide health education. However, utilization of these services is influenced by socio-economic status, level of education, cultural beliefs, and awareness of maternal health issues (GSS, 2021). These contextual factors make Daboya an appropriate setting for examining hypertensive disorders of pregnancy and their associated risk factors.

3.2 Study Design and Type

This study will adopt a descriptive cross-sectional study design, which is widely used in public health research to assess the prevalence and associated factors of health conditions within a defined population at a specific point in time (Creswell & Creswell, 2018).

The cross-sectional design is appropriate because it allows for the simultaneous measurement of the outcome variable (hypertensive disorders of pregnancy) and independent variables such as socio-demographic characteristics, obstetric history, and lifestyle factors. This enables the identification of associations between variables without the need for long-term follow-up (Setia, 2016).

The descriptive nature of the study allows for a detailed characterization of the study population, including their demographic profile, health conditions, and preventive practices. Such designs are particularly useful in epidemiological studies aimed at estimating disease burden and identifying potential risk factors (Levin, 2006).

One of the strengths of the cross-sectional design is its cost-effectiveness and ability to generate data within a short period, making it suitable for resource-limited settings (Creswell & Creswell, 2018). However, a major limitation is that it does not establish causality, as exposure and outcome are assessed simultaneously (Setia, 2016). Despite this limitation, the design is suitable for achieving the objectives of this study.

3.3 Study Population

The study population will consist of pregnant women attending antenatal care services at selected health facilities in Daboya during the study period. Antenatal clinics are considered appropriate settings for such studies because they provide access to pregnant women at different stages of pregnancy (WHO, 2022).

The target population includes all pregnant women residing in Daboya and surrounding communities. However, the accessible population will be limited to those attending antenatal clinics during the data collection period due to logistical constraints.

Inclusion criteria will include pregnant women attending antenatal care, those willing to participate, and those who provide informed consent. Exclusion criteria will include pregnant women who are critically ill or unable to provide reliable responses, as this may compromise data quality (Polit & Beck, 2017).

This population is appropriate because antenatal clinics serve as primary points for screening and managing hypertensive disorders, thereby providing relevant data for the study.

3.4 Sample Size and Sampling Technique

Sample Size Determination

The sample size will be determined using the single population proportion formula, which is widely used in epidemiological research (Cochran, 1977).

$n = (Z^2 \times p \times q) / d^2$

Where:

Z = 1.96 (95% confidence level)

$p = 0.125$ (prevalence from Bugri et al., 2023)

$q = 1 - p = 0.875$

$d = 0.05$ (margin of error)

This gives a sample size of approximately 168 participants. A 10% non-response rate will be added to account for incomplete responses, resulting in a final sample size of approximately 185 respondents. Adjusting for non-response is recommended in survey-based studies to maintain statistical validity (Kish, 1965).

Sampling Technique

A systematic random sampling technique will be used to select participants. This method is appropriate when sampling from a continuous flow of participants, such as antenatal clinic attendees (Etikan & Bala, 2017).

The sampling interval (k) will be calculated by dividing the estimated number of ANC attendees by the sample size. A random starting point will be selected, and every kth participant will be recruited.

This method ensures representativeness and minimizes selection bias. However, care will be taken to ensure that the sampling interval does not coincide with any hidden patterns in attendance (Etikan & Bala, 2017).

3.5 Study Variables

The dependent variable in this study is the presence or absence of hypertensive disorders of pregnancy, determined through clinical diagnosis.

Independent variables include socio-demographic factors (age, education, income), obstetric factors (parity, gestational age), clinical factors (family history, pre-existing conditions), and lifestyle factors (diet, physical activity). These variables have been identified in previous studies as significant determinants of hypertensive disorders (WHO, 2022; Gemechu et al., 2020).

3.6 Data Collection Tool and Technique

Data will be collected using a structured questionnaire and review of antenatal records. Structured questionnaires are widely used in health research due to their ability to standardize data collection and improve reliability (Polit & Beck, 2017).

Face-to-face interviews will be conducted to ensure clarity and inclusiveness, particularly in populations with low literacy levels. Clinical data will be extracted from antenatal records to improve accuracy and reduce recall bias (WHO, 2022).

A pre-test will be conducted to assess the validity and reliability of the instrument. Pre-testing is essential in identifying ambiguities and ensuring consistency in responses (Creswell & Creswell, 2018).

3.7 Data Processing and Analysis

Data will be coded and entered into SPSS version 25 for analysis. SPSS is widely used in health research due to its efficiency in handling large datasets (Field, 2018).

Descriptive statistics will summarize the data, while inferential statistics such as Chi-square tests and logistic regression will assess associations between variables. Logistic regression is particularly useful for identifying predictors while controlling for confounding variables (Hosmer et al., 2013).

Statistical significance will be set at $p < 0.05$, which is a standard threshold in epidemiological research (Field, 2018).

3.8 Limitation of the Study

The cross-sectional design limits causal inference, as recommended in epidemiological studies (Setia, 2016). Self-reported data may introduce recall bias, and facility-based sampling may limit generalizability (Polit & Beck, 2017).

3.9 Ethical Considerations

Ethical approval will be obtained in line with international research standards involving human subjects (WHO, 2011). Informed consent, confidentiality, and voluntary participation will be strictly upheld.

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